



University of
Nottingham

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COMP4139

Machine Learning (MLE)

- Method Evaluation

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- Neural networks – one line?
- Different layers in a neural network
- Activation Functions – one line?
- Loss Functions – one line?
- Backpropagation – one line?
- Example of backpropagation



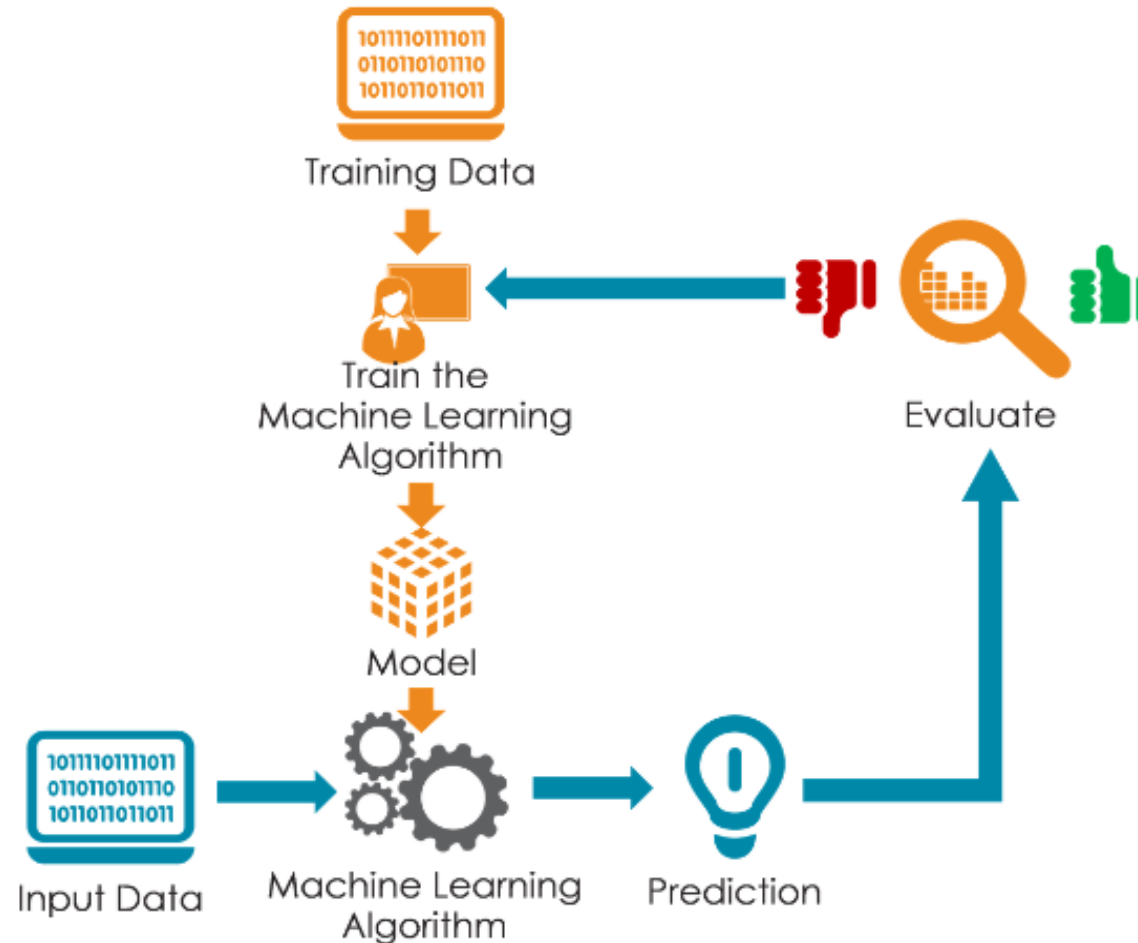
What you will learn in this lecture

- How do we evaluate ML models?
- What are some commonly used metrics?
- How do we know if the results we have got are statistically significant?



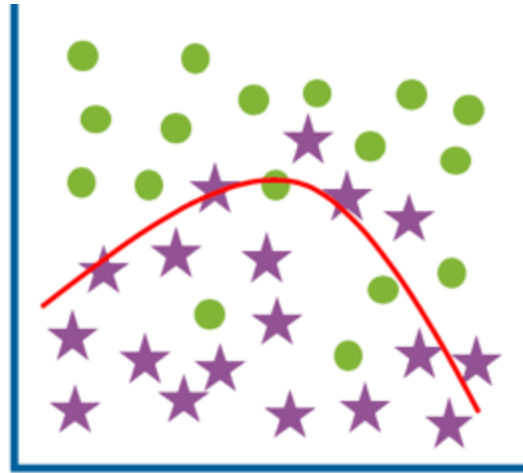
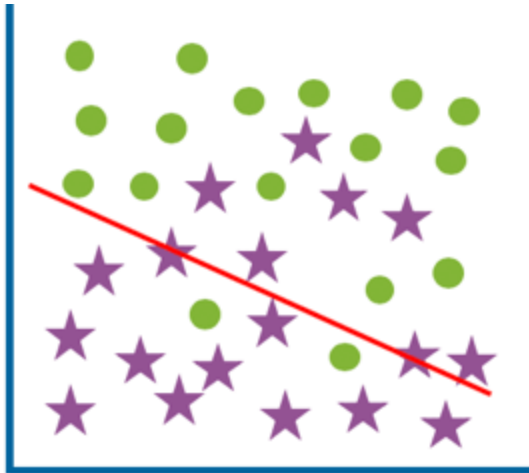
General Pipeline of Machine Learning

- Data representation + Modelling + Evaluation + Optimisation (Pedro Domingos, 2012)





Method Generalisation





Method Generalisation

- **Generalisation** is the desired property of a ML model to be able to predict outcome of unseen examples correctly.
- A hypothesis generalises well if it can predict an example coming from the same distribution as the training examples well.
- A hypothesis is said to be **overfit** if its prediction performance on the training data is **overoptimistic** compared to that on unseen data. It presents itself in complicated decision boundaries that depend strongly on individual training examples.
- How to measure generalisation error?
 - Training/Validation/Test
 - Cross validation
 - Independent test set



K-fold Cross Validation

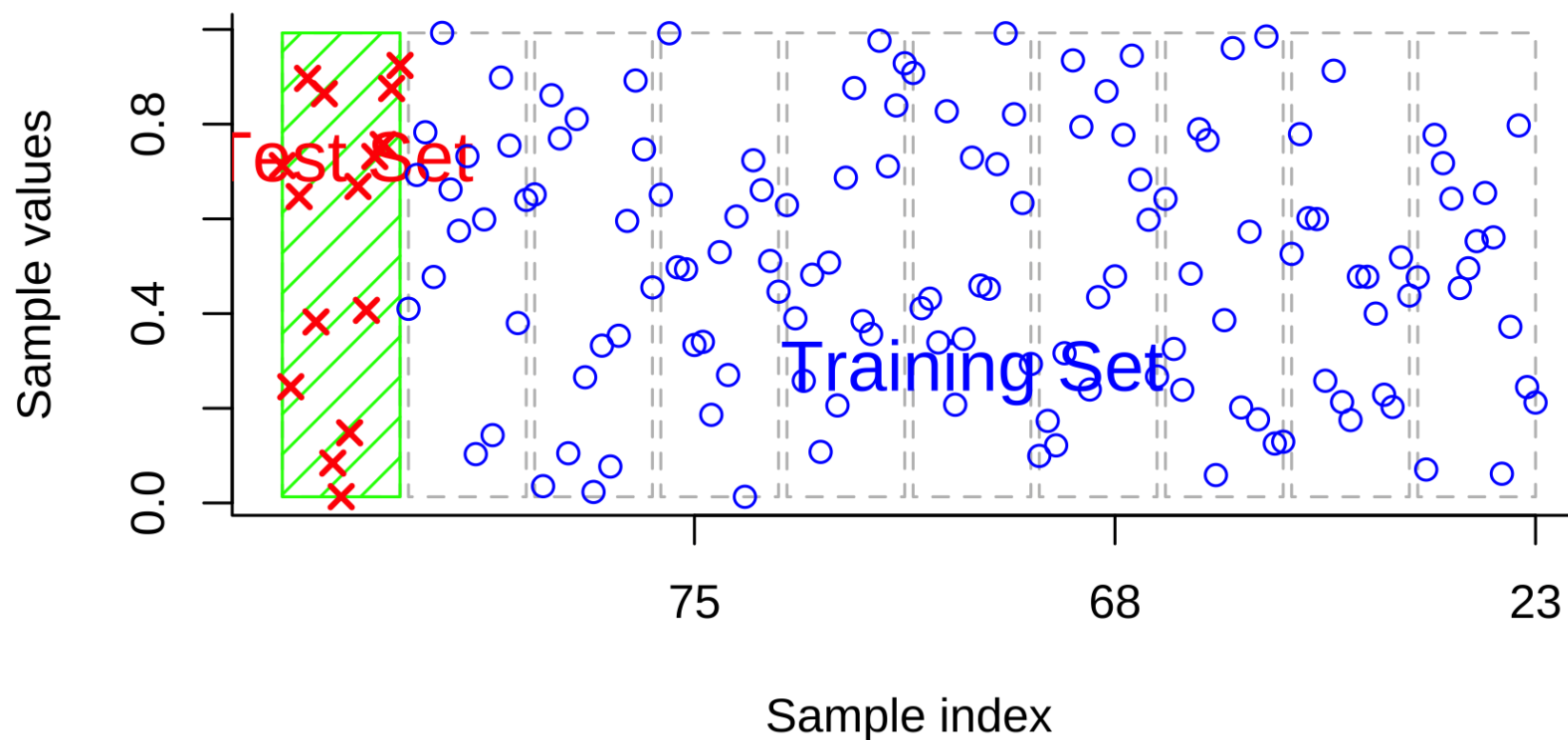
- Split training data into K folds.
- For each fold, train on all other folds and make predictions on the held-out test fold.
- Iterate all folds as the test fold.
- Combine all the results of K folds, so that every data point has been used as a test data point.
- How do we tune the hyperparameters?





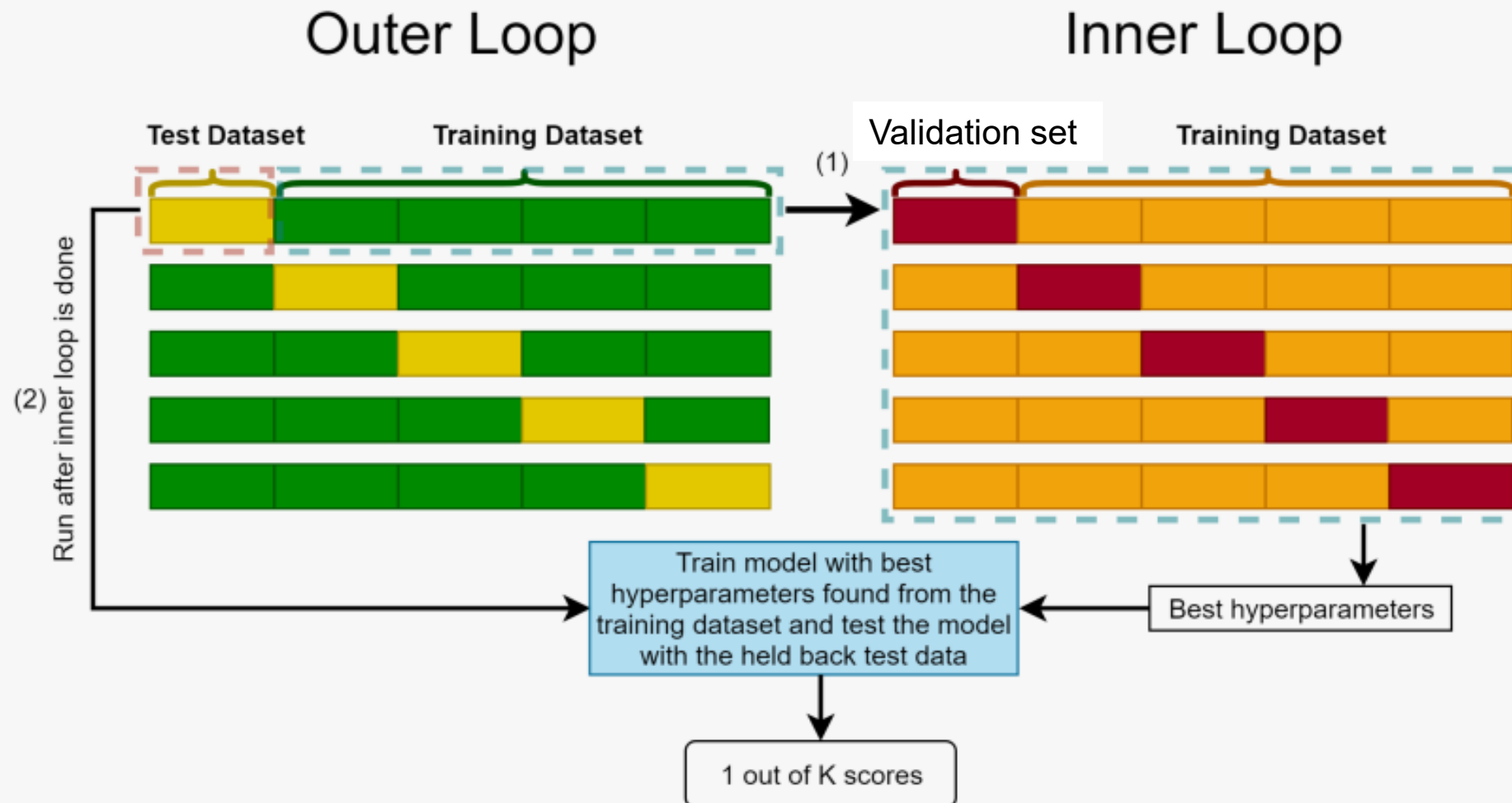
K-fold Cross Validation

Demonstration of 10-fold Cross Validation



Nested K-fold Cross Validation

In the inner-loop, the **same** set of hyperparameters are used in all folds.





TP, FP, TN, FN





Evaluation Metrics (Classification)

- True positive, false positive, true negative, false negative

	Predicted Positive	Predicted Negative
Actual Positive	TP	FN
Actual Negative	FP	TN

- Classification accuracy= $(TP+TN)/(TP+FP+FN+TN)$,
- Suppose we build a model to detect diabetes. We test 1000 patients, only 50 have the disease.

	Predicted Positive	Predicted Negative
Actual Positive	10	40
Actual Negative	10	940

- Classification accuracy= 95%
- Good or bad model?



Precision, Recall, Sensitivity and Specificity

		Predicted			
		Predicted Positives TP+FP	Predicted Negatives TN+FN		
Real (Actual, Observed)	Real Positives TP+FN	true positives (TP)	false negatives (FN)	Sensitivity SNIP (SeNsitivity Is Positive) true positives/real positives $TP/(TP+FN)$	Recall true positives/ REAL positives $TP/(TP+FN)$ Recall = Sensitivity
	Real Negatives TN+FP	false positives (FP)	true negatives (TN)	Specificity SPIN (SPecificity Is Negative) true negatives/real negatives $TN/(TN+FP)$	
		Precision true positives/ PREDiCted positives $TP/(TP+FP)$		Accuracy true predictions/all predictions $(TP+TN)/(TP+TN+FP+FN)$	

Evaluation Metrics (Classification)

- True positive, false positive, true negative, false negative

	Predicted Positive	Predicted Negative
Actual Positive	TP	FN
Actual Negative	FP	TN

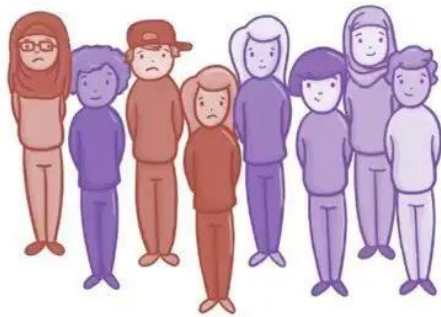
- Classification accuracy= $(TP+TN)/(TP+FP+FN+TN)$,
Balanced classification accuracy= $(\text{Sensitivity} + \text{Specificity}) / 2$
- Precision= $TP/(TP+FP)$ and Recall = $TP/(TP+FN)$
 - Precision: percentage of correct positive predictions among all predicted positive predictions.
 - Recall: percentage of correct positive predictions among all real positive cases. (e.g. don't want to miss any diseased cases)
- Sensitivity=Recall and Specificity= $TN/(TN+FP)$
 - Specificity: percentage of correct negative predictions among all real negative cases.



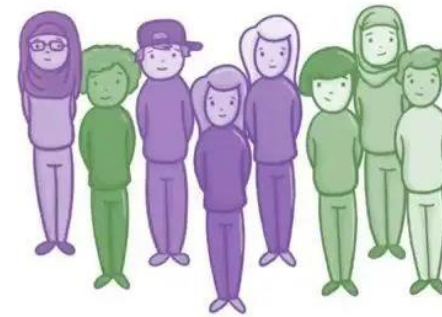
Sensitivity vs Specificity

CAN the TEST CORRECTLY IDENTIFY
if a PERSON has DIABETES or NOT ?
= **VALIDITY**

NEW SCREENING TEST



IDENTIFIES MOST PEOPLE
WITH the **CONDITION**



IDENTIFIES MOST PEOPLE
WHO DON'T HAVE
the **CONDITION**

Evaluation Metrics (Classification)

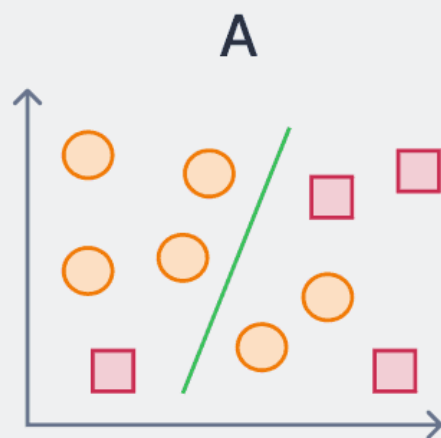
- True positive, false positive, true negative, false negative

	Predicted Positive	Predicted Negative
Actual Positive	10	40
Actual Negative	10	940

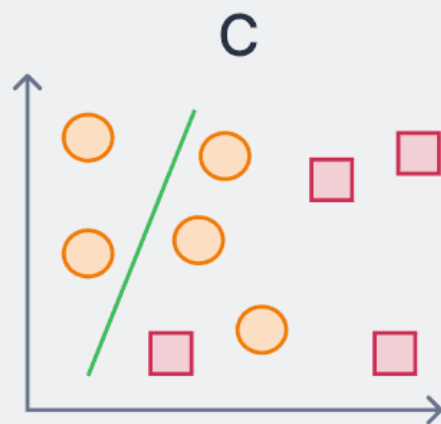
- Classification accuracy= $(TP+TN)/(TP+FP+FN+TN) = 95\%$
- Precision= $TP/(TP+FP) = 50\%$ and Recall = $TP/(TP+FN) = 20\%$
 - Precision: percentage of correct positive predictions among all predicted positive predictions. Only half of positive predictions are correct.
 - Recall: percentage of correct positive predictions among all real positive cases. (e.g. don't want to miss any diseased cases). The model misses 80% of real patients!
- Sensitivity=Recall and Specificity= $TN/(TN+FP) = 98.9\%$
 - Specificity: percentage of correct negative predictions among all real negative cases. Great at identifying healthy patients.



Precision vs Recall

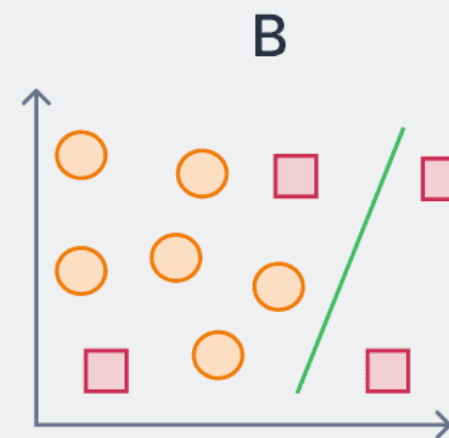


Increase
Recall



P = 50%; R = 100%

Increase
Precision



P = 100%; R = 50%

□ → Positive class

○ → Negative class

/ → Model decision
boundary

Negative class / Positive class



- Accuracy is fine when dataset is balanced.
- Both precision and recall are useful metrics in cases where the dataset is imbalanced.
- COVID-19 -> avoid spreading, what is more important FP or FN?
- In this use case, false positives (a healthy patient diagnosed as COVID-positive) are not as important as preventing a contagious patient from spreading the disease.
- In most high-risk disease detection cases (like cancer), recall is a more important evaluation metric than precision.
- However, precision is more useful when we want to affirm the correctness of our model. Ex. Instagram or YouTube recommendations.
- Think of precision as "how many of these are relevant?" and recall as "how many of the relevant items did we find?"



F Measure

- Comparing different approaches is difficult when using multiple evaluation measures (e.g. Recall and Precision)
- F measure combines recall and precision into a single measure.
- F measure ranges from 0 to 1. 1 indicates perfect precision and recall.
- F_1 is commonly used. Beta=1, P: Precision, R: Recall
- Beta could make either precision or recall more important.

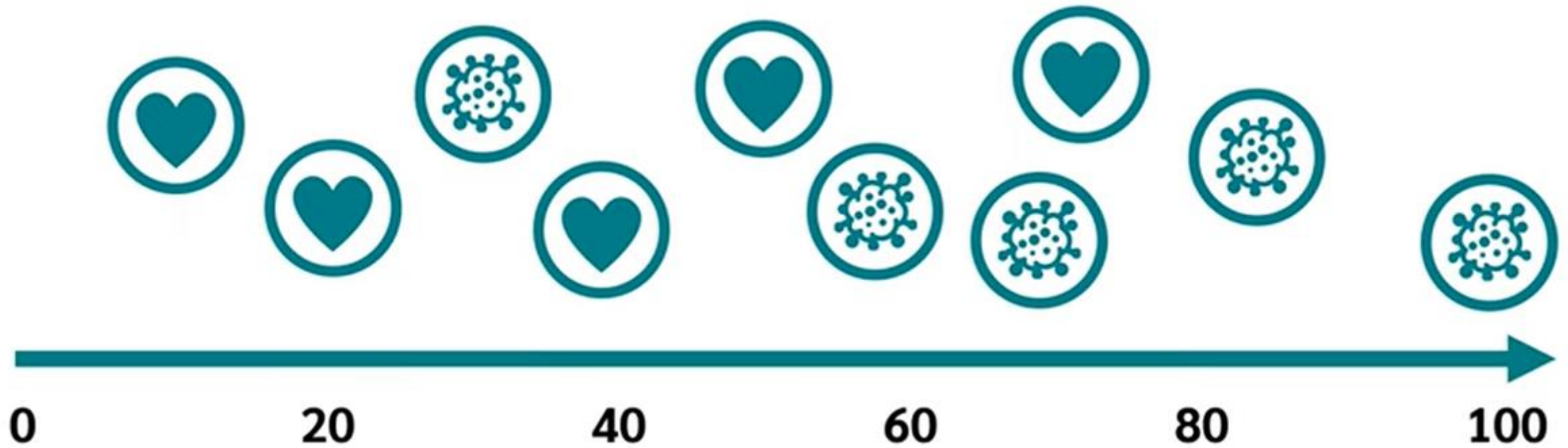
$$f_{\beta} = (1 + \beta^2) \frac{PR}{(\beta^2 P) + R}$$

$$F_1 = 2 \cdot \frac{0.50 \times 0.20}{0.50 + 0.20}$$

$$F_1 = 2 \cdot \frac{0.10}{0.70} = 0.2857$$



TPR and FPR





TPR and FPR

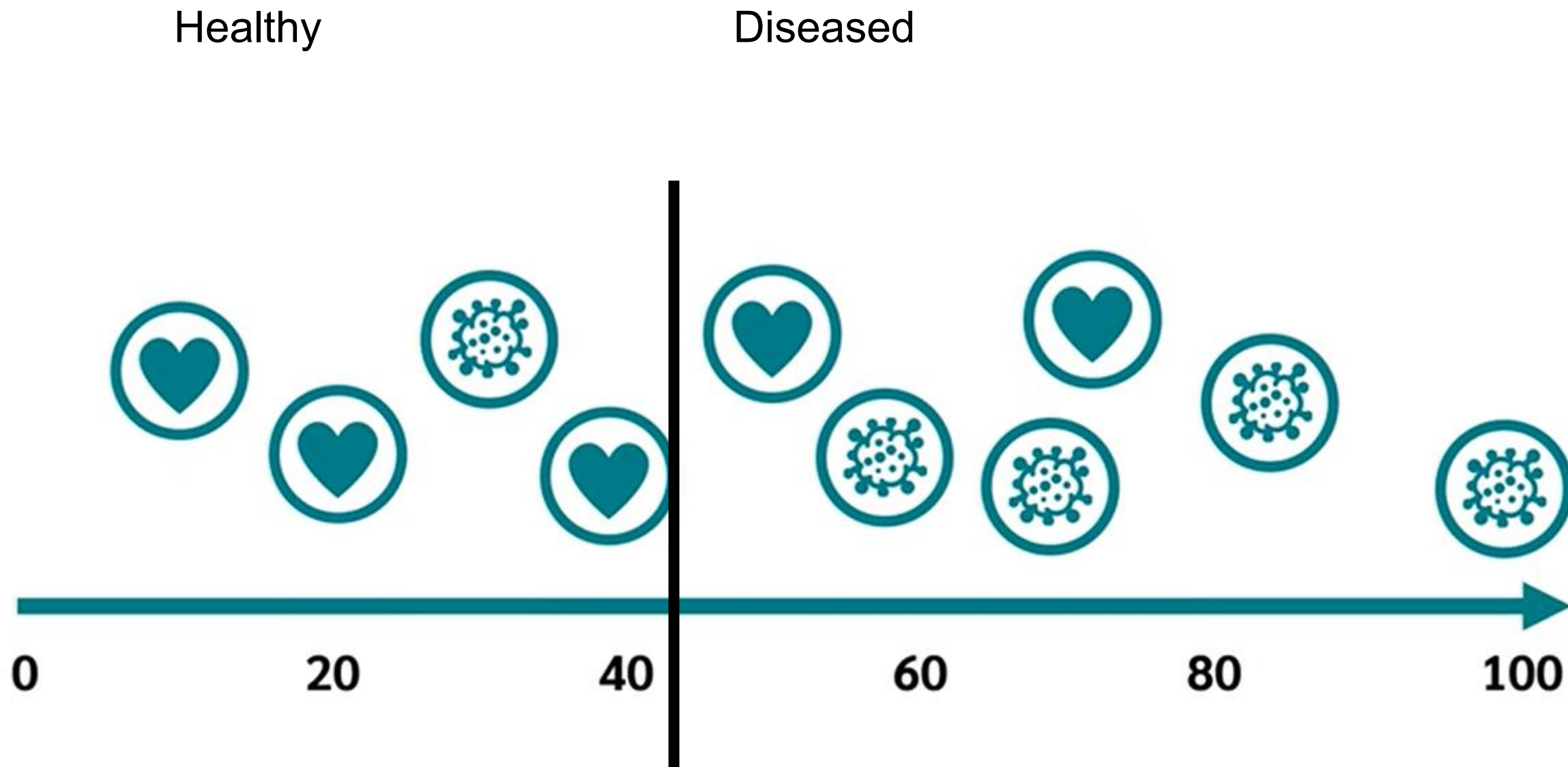
Actual	0	TN	FP
	1	FN	TP
		0	1
		Predicted	

*False Positive
Rate (FPR) = $\frac{FP}{(FP + TN)}$*

*True Positive
Rate (TPR) = $\frac{TP}{(TP + FN)}$*

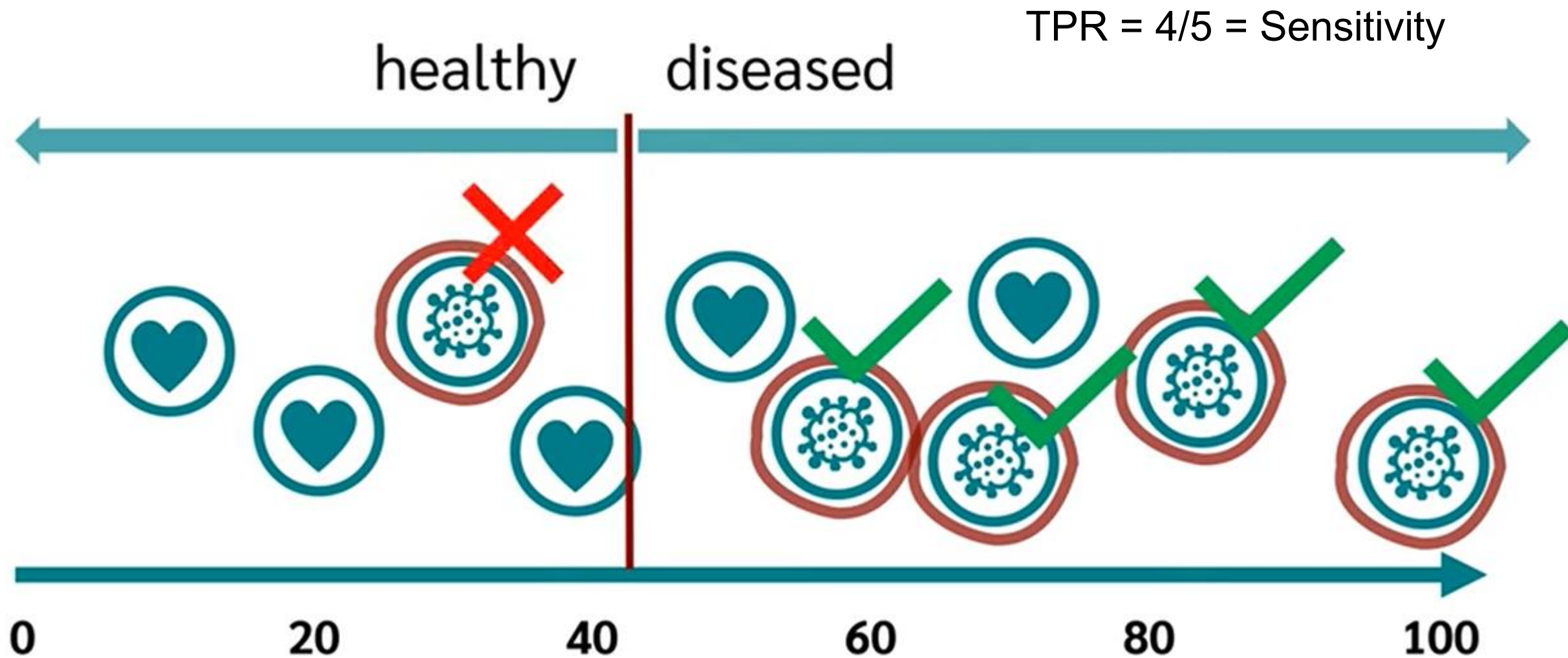


TPR and FPR





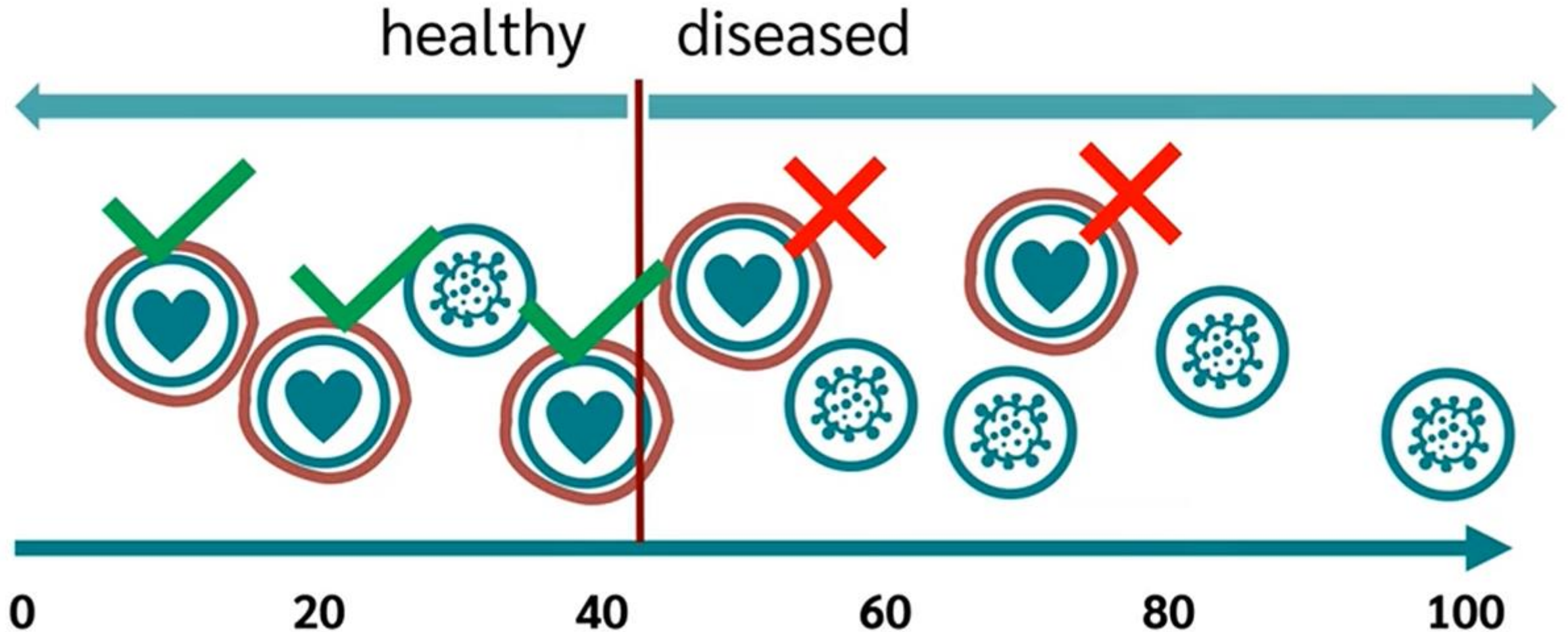
TPR and FPR





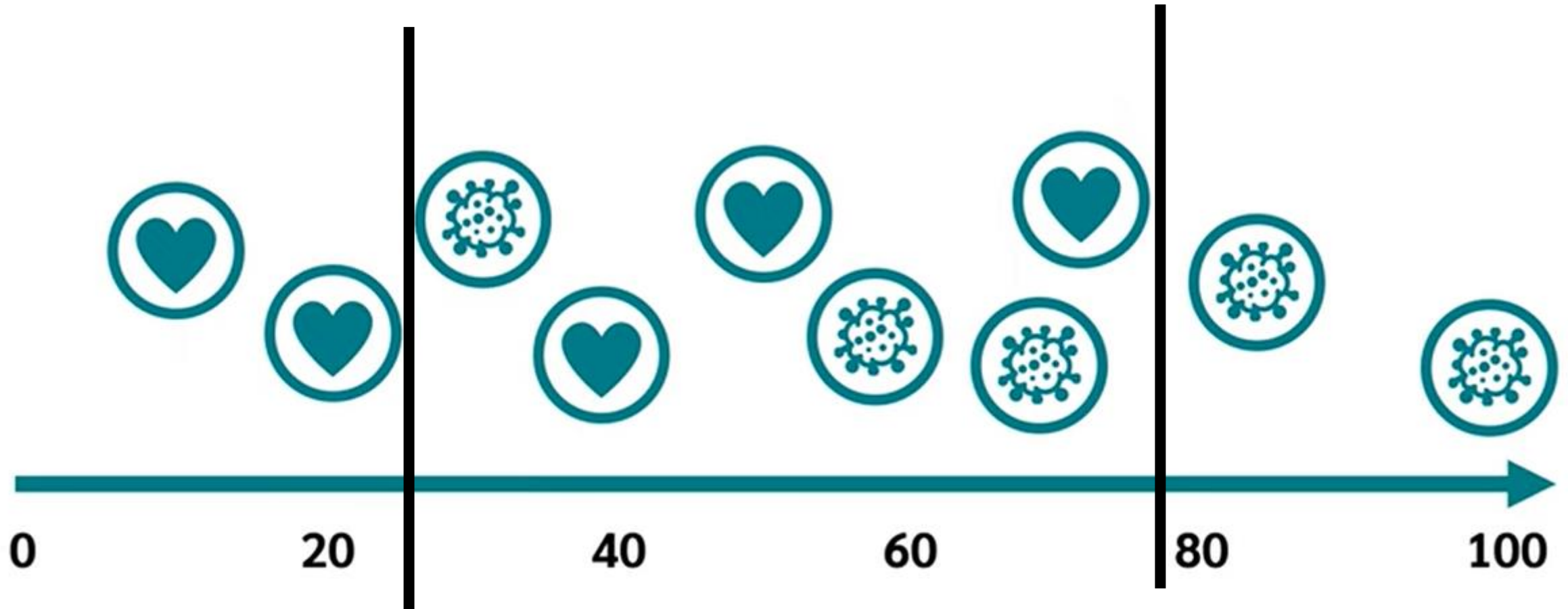
TPR and FPR

$$\text{FPR} = 2/5 = 1 - \text{Specificity}$$





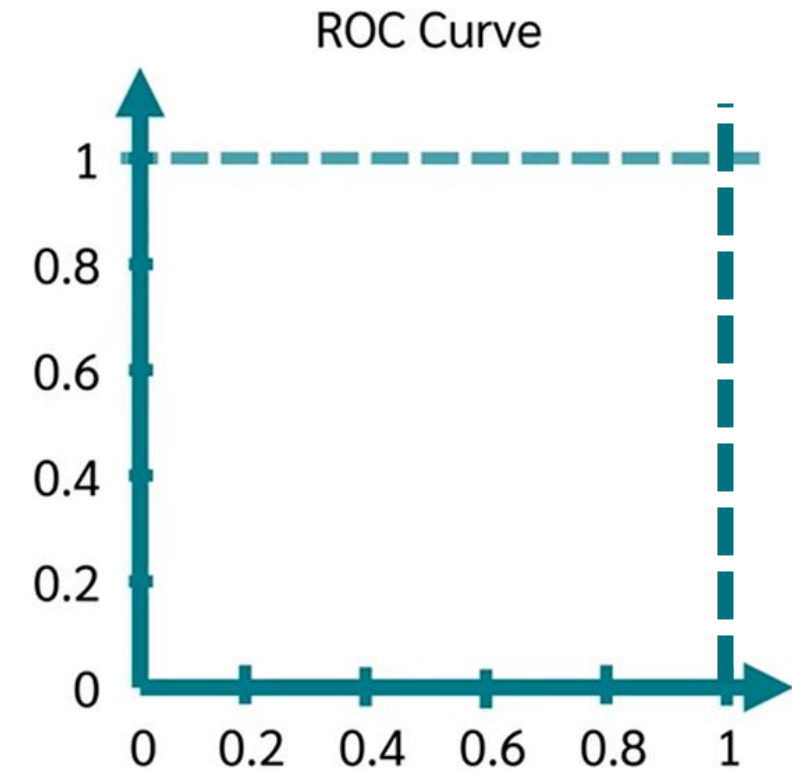
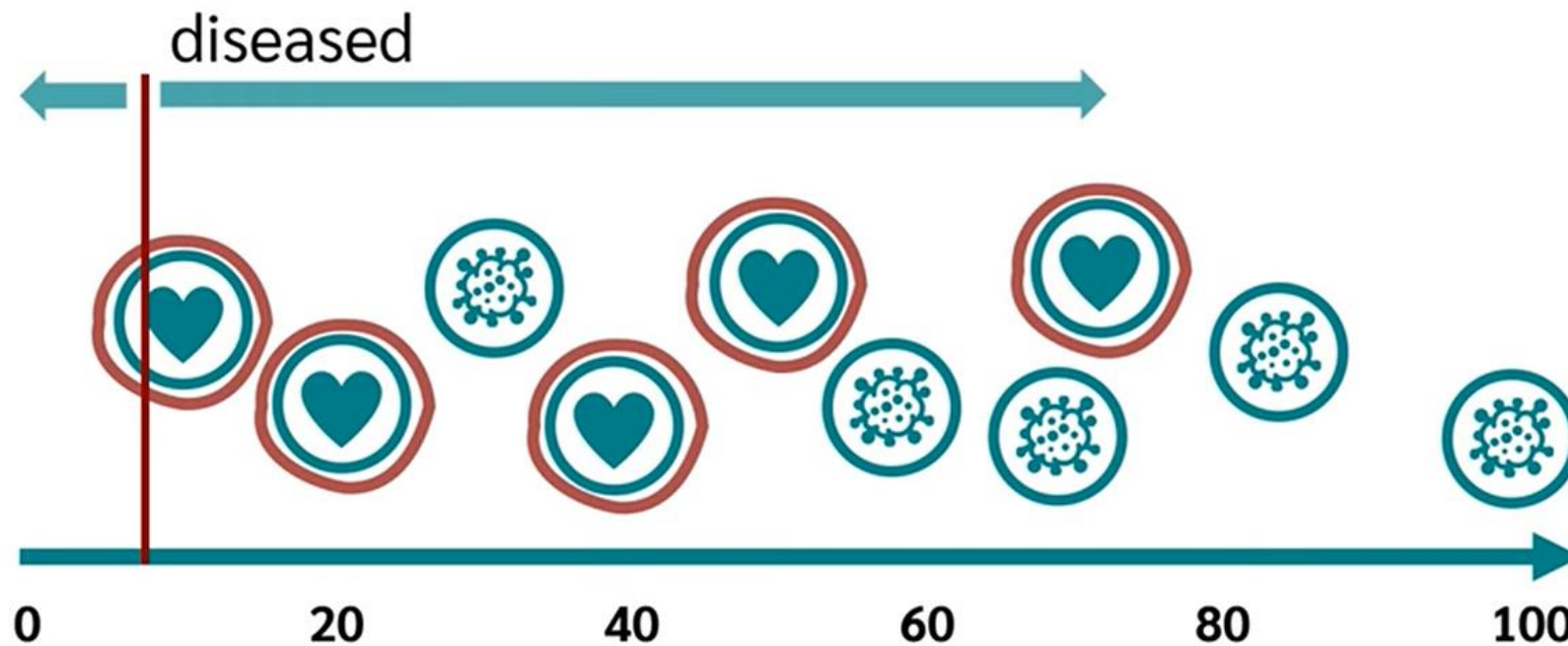
TPR and FPR





TPR and FPR

$$\text{TPR} = 5/5 = \text{FPR}$$

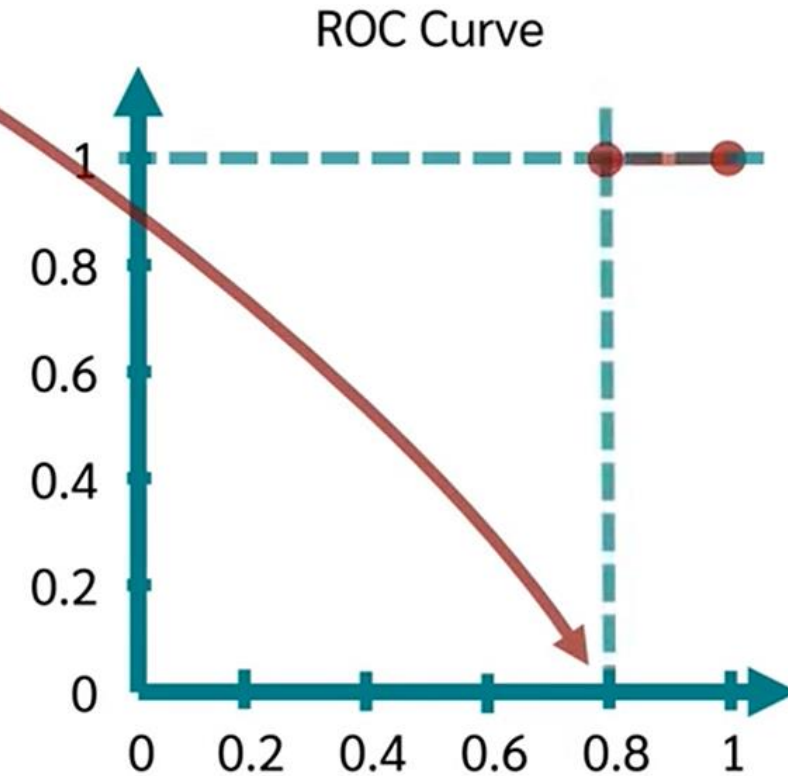
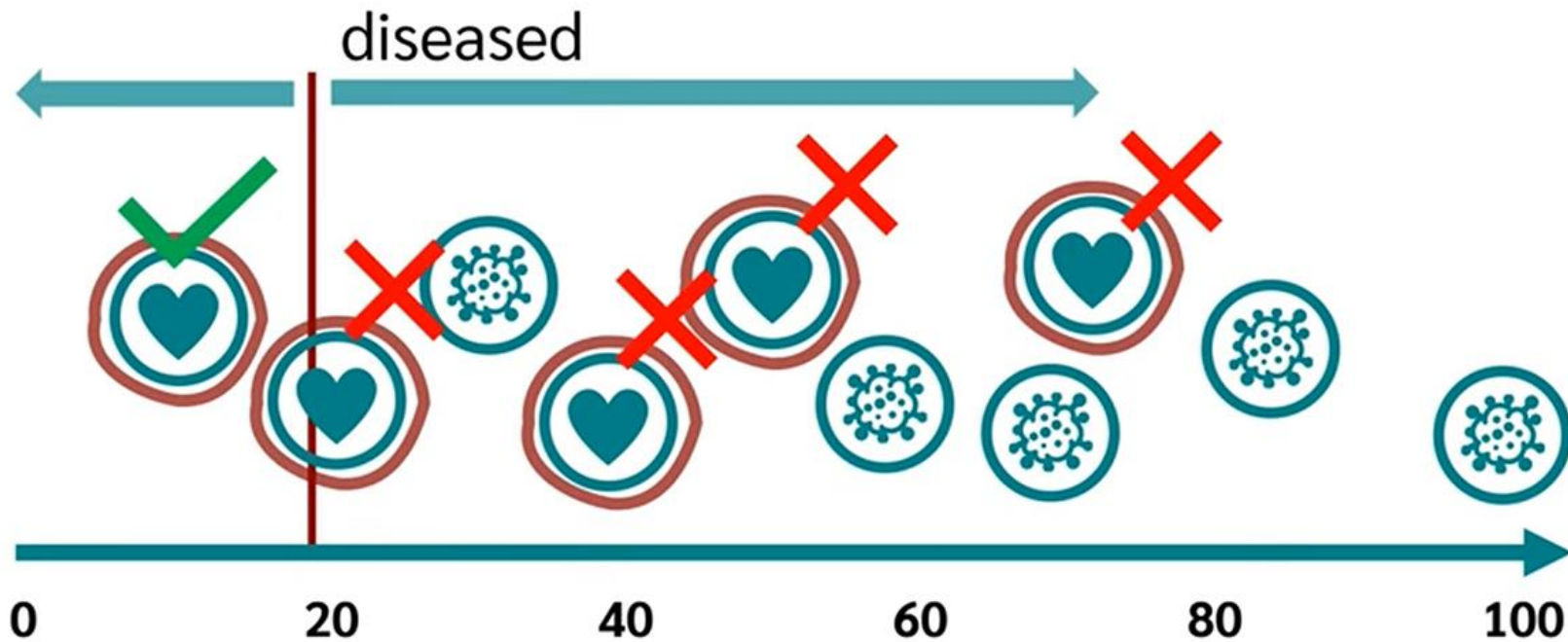


TPR and FPR

$$\text{TPR} = 5/5$$

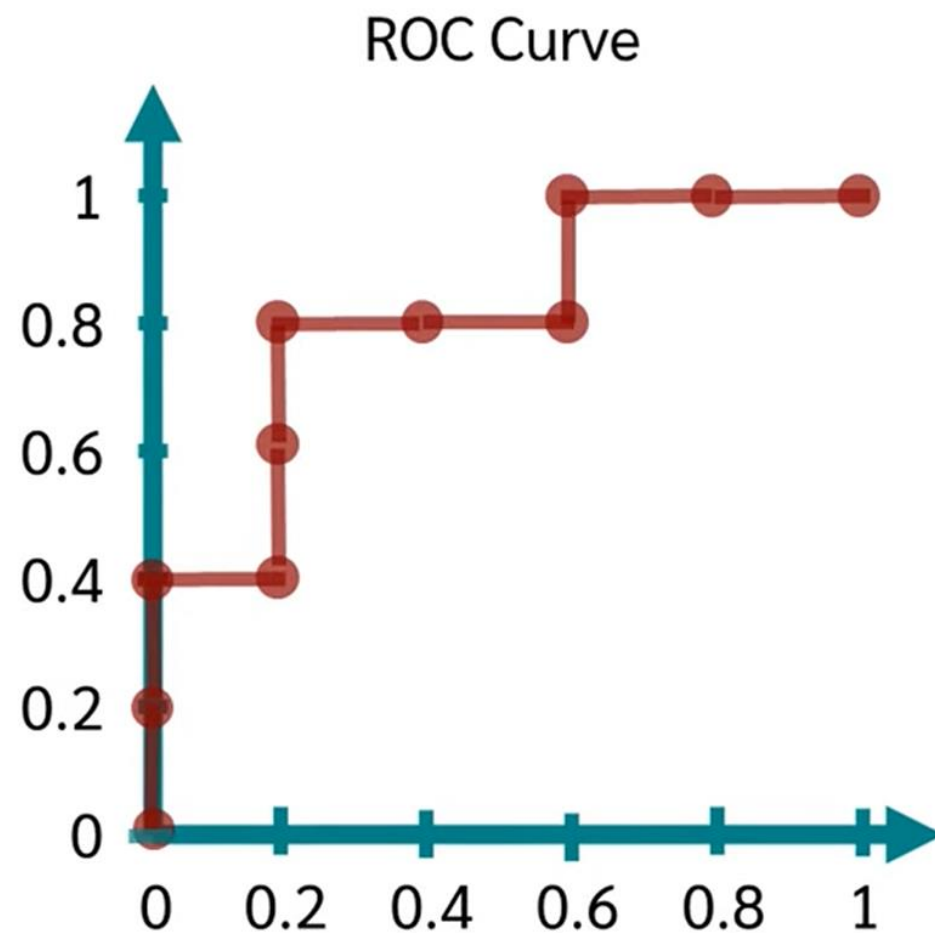
$$\text{FPR} = 4/5$$

So, we have **4** out of **5** or **0.8**.



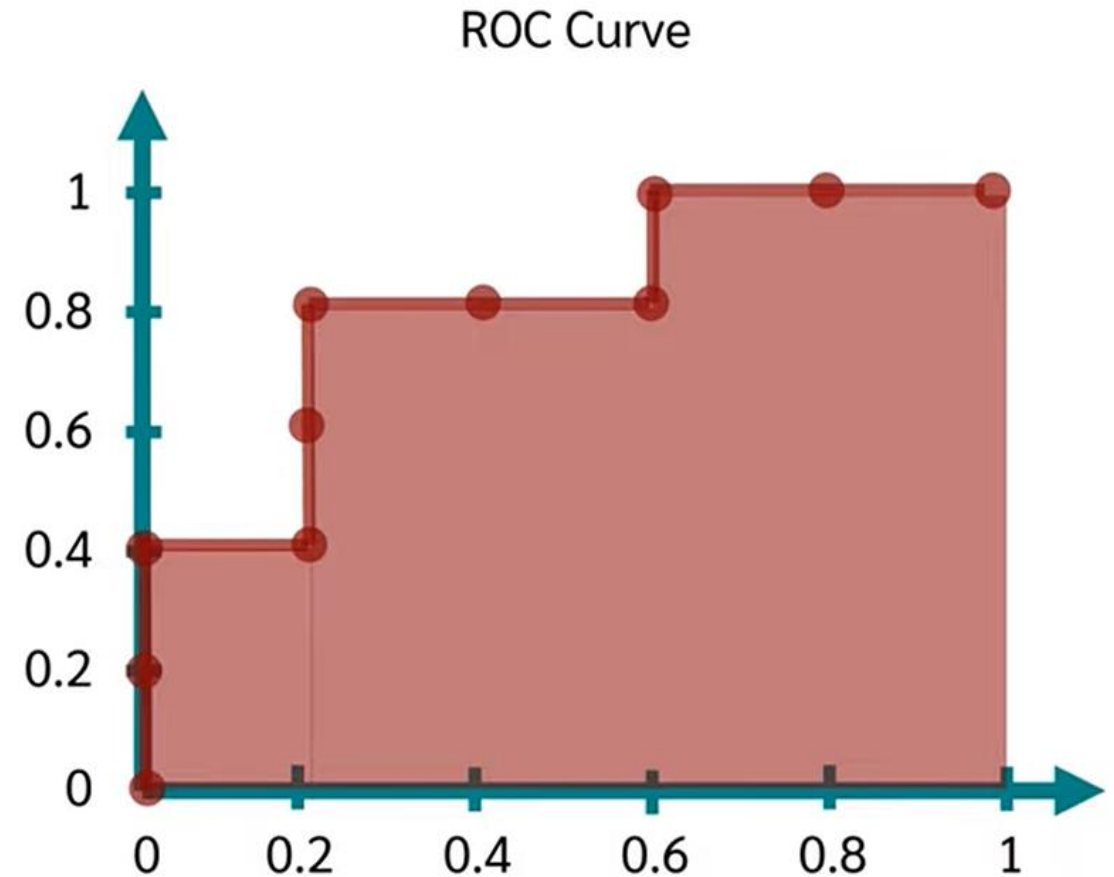


TPR and FPR





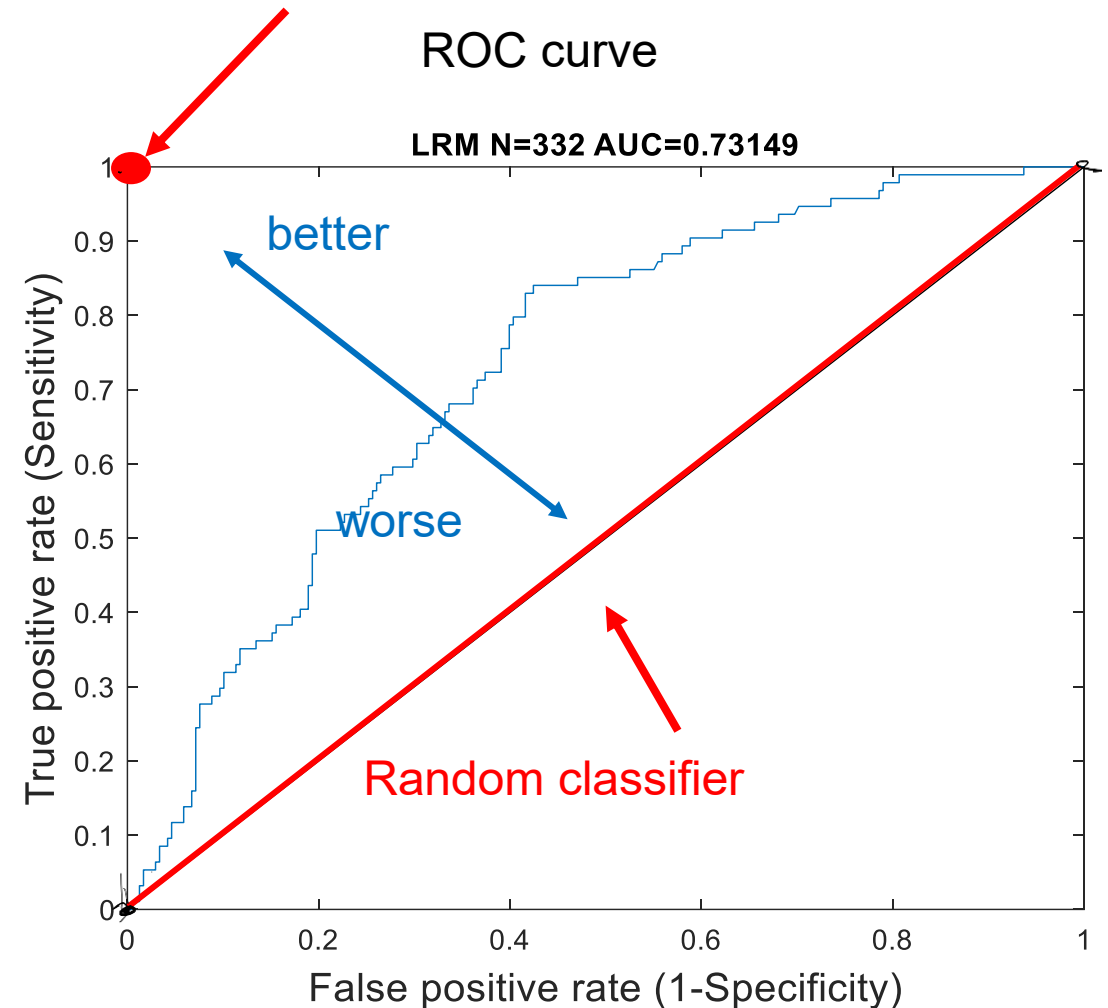
Therefore, the larger the
area under the curve,
the better the classifier is.



Receiver Operator Characteristic (ROC) Curve

- Calculated by varying decision threshold of a classifier
- Area under the ROC curve (AUROC or AUC) is a widely used measure of goodness

Perfect classifier



- A visualisation tool used to present the results attained by a learner.
- Easy to see if the system is commonly mislabelling one class as another.

	Predicted Positive	Predicted Negative
Actual Positive	TP	FN
Actual Negative	FP	TN

Binary Case

Actual	Elephant	Monkey	Fish	Lion
	25	3	0	2
	3	53	2	3
	2	1	24	2
	1	0	2	71
	Elephant	Monkey	Fish	Lion
Predicted				

Multi-class Case

Evaluation Metrics (Regression)

- Mean Squared Error

- $MSE = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$

$\hat{Y}$: predicted values, Y : observed values,
n: number of data samples

- The squaring has the effect of inflating or magnifying large errors.

- Root Mean Squared Error

- $RMSE = \sqrt{MSE}$

- Punish more on large errors but retains the original unit of the target value.

- Mean Absolute Error

- $MAE = \frac{\sum_{i=1}^n |Y_i - \hat{Y}_i|}{n}$

- Retains the original unit of the target value, and the changes are linear.

Statistical Tests (Comparing two methods)

Fold	SVM Accuracy	ANN Accuracy
1	0.82	0.85
2	0.80	0.88
3	0.79	0.84
4	0.83	0.87
5	0.81	0.86
6	0.82	0.89
7	0.78	0.84
8	0.80	0.88
9	0.81	0.86
10	0.79	0.87

[t-tests and p values](#)

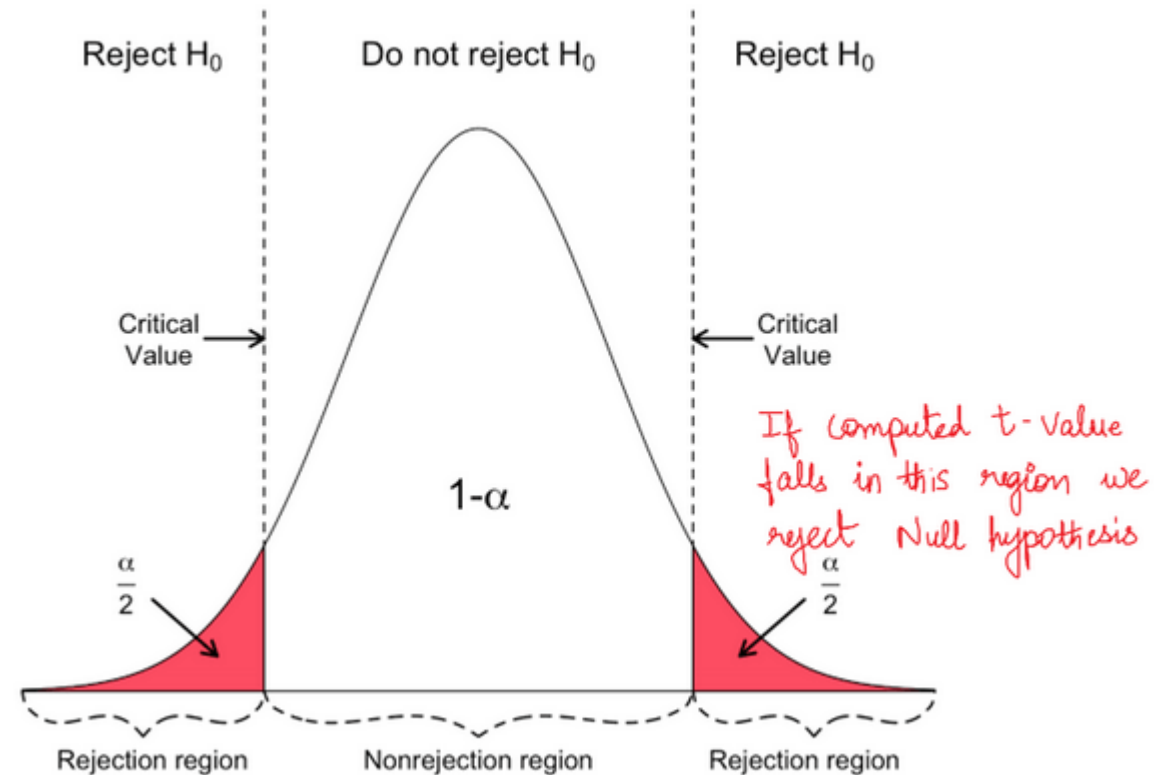
Statistical Tests (Comparing two methods)

Hypothesis: ANN is better than SVM

Null Hypothesis: They are the same on average

T-test (parametric)

- How the **mean** of two groups are different from each other?
- **Assumes normal distribution.**
- null-hypothesis: the mean of the two groups are the same.
- If $p < 0.05$, it is statistically significant.
- Larger sample size is more likely to be statistically significant.



t-tests and p values



p-value

Descriptive Statistics				
N	Mean	StDev	SE Mean	95% CI for μ
30	10.096	0.799	0.146	(9.798, 10.394)
μ : mean of Cycle time				
Test				
Null hypothesis		$H_0: \mu = 10.1$		
Alternative hypothesis		$H_1: \mu \neq 10.1$		
T-Value	P-Value			
-0.03	0.979			

The P-value is 0.979 meaning it is 97.9% likelihood that mean 10.1 will be within the 95% confidence interval of 9.798 and 10.394

Descriptive Statistics				
N	Mean	StDev	SE Mean	95% CI for μ
30	10.096	0.799	0.146	(9.798, 10.394)
μ : mean of Cycle time				
Test				
Null hypothesis		$H_0: \mu = 10.5$		
Alternative hypothesis		$H_1: \mu \neq 10.5$		
T-Value	P-Value			
-2.77	0.010			

The P-value is 0.01 meaning there is a 1% chance that mean 10.5 will fall within the interval of 9.798 and 10.394.

Descriptive Statistics				
N	Mean	StDev	SE Mean	95% CI for μ
30	10.096	0.799	0.146	(9.798, 10.394)
μ : mean of Cycle time				
Test				
Null hypothesis		$H_0: \mu = 10.394$		
Alternative hypothesis		$H_1: \mu \neq 10.394$		
T-Value	P-Value			
-2.04	0.050			

The P-value is 0.05 meaning there is a 5% chance that mean 10.394 will fall within the interval of 9.798 and 10.394



1.



One sample
t-test

2.

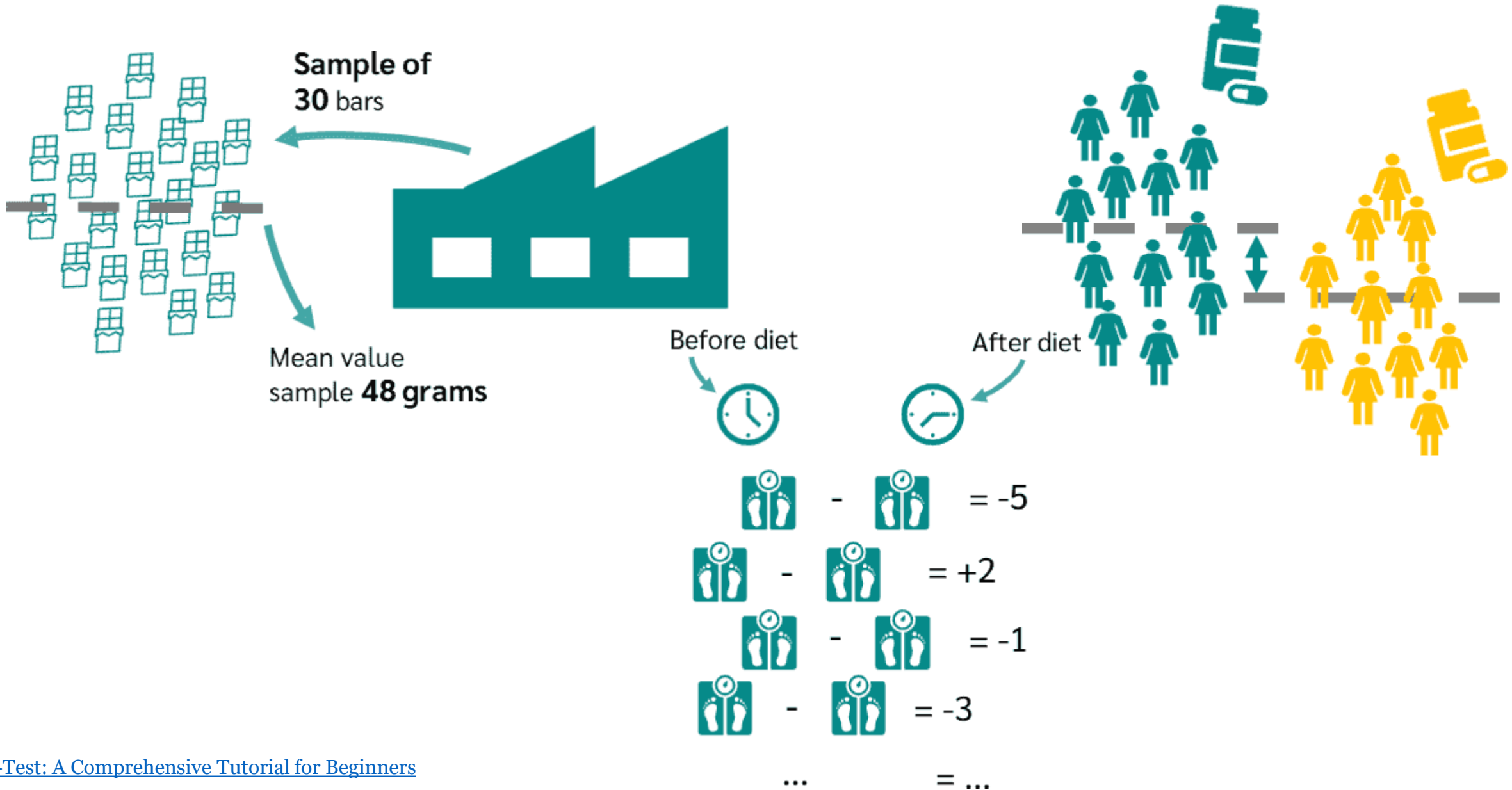


Independent
samples t-test

3.



Paired
samples t-test





Assumption	What it means	Why it matters
1. Normality	The differences between paired observations are normally distributed.	In our SVM vs ANN example, the fold-wise accuracy differences (e.g., 0.03, 0.05, 0.07, ...) should roughly follow a bell-shaped curve. If they don't (e.g., skewed or outliers), the test might misestimate significance.
2. Independence	Each observation (e.g., each fold's result) is independent of the others.	Fold 1 accuracy shouldn't affect Fold 2. If folds overlap or are correlated, the T-test's math breaks down.
3. Continuous/interval data	The data must be numeric and continuous (not categorical).	Accuracy values like 0.82, 0.88 are fine, but "good"/"bad" labels wouldn't work.
4. Equal variances (for <i>independent</i> samples)	The two groups have similar spread (variance).	For a paired T-test (like our case), this assumption isn't needed, we only care about the distribution of <i>differences</i> .

Statistical Tests (Comparing two methods)

Wilcoxon signed-rank test (**non-parametric**):

- How the **median** of two groups are different from each other?
- **No model assumption. Applied to paired data or unpaired data.**
- null-hypothesis: the median of the two groups are the same.
- If $p < 0.05$, it is statistically significant.
- Larger sample size is more likely to be statistically significant.
- Safer to use than T-test.
- Less likely to show statistical significance than T-test, if the data follows normal distribution and paired (use T-test).



Comparing the two

Aspect	Paired T-test	Wilcoxon Signed-Rank Test
Type	Parametric	Non-parametric
Compares	Mean differences	Median/ranked differences
Assumes normality?	✓ Yes	✗ No
Sensitive to outliers?	✓ Yes	✗ No
Uses	Actual values	Rank order of differences
Result here	$p < 0.001$ (significant)	$p < 0.05$ (significant)
Conclusion	ANN significantly better	ANN significantly better



Confidence Interval

- Our sampled data is never enough. CI helps to indicate how confident we are of our estimation.
- A confidence interval is the mean of the estimate plus and minus the variation in that estimate.
- If the data follows a normal distribution, or a large sample size ($n > 30$) that is approximately normally distributed, the z-distribution can be used to find the critical values.

Confidence Level	90%	95%	99%
Z score	1.64	1.96	2.57

- Bootstrapping: random sampling with replacement. Calculate the mean and standard deviation, which is applied to the testing process.

$$CI = \bar{X} \pm Z \times \frac{\sigma}{\sqrt{n}}$$

Handwritten annotations:

- mean (pointing to $\bar{X}$)
- standard deviation (pointing to σ)
- sample size (pointing to n)



Recap

- How do we evaluate ML models?
- What is overfitting and underfitting?
- What are some commonly used metrics?
- How do we know if the results we have got are statistically significant?



What we will learn in the next lecture

- What is unsupervised learning?
- Similarity between vectors
- K-Means Clustering
- Agglomerative Hierarchical Clustering
- Density-based Spatial Clustering of Applications with Noise (DBSCAN)
- Expectation Maximisation